AN ETHICAL AND EMOTIONALLY INTELLIGENT SOCIAL MEDIA PLATFORM USING RESPONSIBLE AND EXPLAINABLE AI

Project Id: 25-26J-137

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Computer Science and Software Engineering

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor:

(Mr. Samadhi Rathnayake) Date: 28/8/25

ABSTRACT

This proposal introduces the component **Cyberbullying Detection and Behavior Profiling of Harmful Users** which will be based on machine learning. It is a part of a social media platform that aims to act as an ethical and emotionally intelligent platform. This component considers both the victims and the offenders views. NLP (Natural Language Processing) techniques will be used to analyze negative textual contents such as comments. Transformer-based models and Explainable AI (XAI) will be used to give reasons to why contents were considered as Harmful Bullying content. Instead of relying on User manual reports, this system tracks user behavior overtime and flags harmful users and content mainly based on the frequency of posting harmful posts. The system will detect harmful posts and automatically delete and explain why the content were considered as harmful. This approach will protect the victim and also gives the offender a sense of transparency regarding why their content were deleted. Further this component will detect negative image-based content such as memes to build a more safe and emotionally intelligent platform that align with ethical AI principles. This component is a sub part of a larger platform and it will be integrated with three other machine learning-based components, Emotion-Aware Reaction System & Ethical Search Filtering, Explainable Friend Recommendation Engine and Behavior-Based Fake Account Detection. This system combines features that results in building a social media platform minimizing cyberbullying and creates a safer and emotionally intelligent environment for users.

Key words: Cyberbullying Detection, NLP (Natural Language Processing), Machine Learning, Explainable AI (XAI), Transformer-based models.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
CNN	Convolutional Neural Network
MongoDB	Document-Oriented NoSQL Database
NLP	Natural Language Processing
REST	Representational State Transfer
RoBERTa	Robustly Optimized BERT Pretraining Approach
SVM	Support Vector Machine
UAT	User Acceptance Testing
XAI	Explainable Artificial Intelligence

1. INTRODUCTION

In Todays' era Social Media Platform are being used by worldwide users and it has become an important tool for users. They use it for business purposes, sharing information, self-expression and a tool for communication. While these platforms provide many benefits for people, there are many drawbacks of using social media platforms as well. Cyberbullying is one such disadvantage. Due to Cyberbullying social media can be an unsafe environment for users. And it may influence negatively towards their mental wellbeing. With the evolution of the technology and with number of users increasing day by day, cyber bullying over social networks has become an increasing problem. And the number of fake accounts have also drastically increased which can undermine trust on social networks.

To handle these issues, our project aims to design and develop an ethical and emotionally intelligent social media platform that provides the user a safe environment. The social media platform will be built machine learning- based and the use of responsible and explainable AI will provide user with the transparency in the operations performed. The system consists of four AI-driven components, focusing on social media challenges faced by users. The components are:

Cyberbullying Detection & Behavior Profiling: This component automatically detects and delete harmful content and tracks users posting patterns and flags them.

Emotion-Aware Reaction System & Ethical Search Filtering: this component detects emotional context, prevent misuse of reactions based on the detected emotion and restrict harmful user queries.

Explainable Friend Recommendation Engine: This component suggests new connections to the users based on their various shared interests and explains why the recommendation was made.

Behavior-Based Fake Account Detection: Detect bot-like users using unsupervised machine learning and limit access.

As mentioned above Cyberbullying has become a critical problem that affects negatively towards users. Due to Cyberbullying users may psychologically suffer from various issues such as low self-esteem, being insecure to share valuable information online etc. Therefore, it is important to minimize cyberbullying. However, with a large number of users using social media it is difficult to manually analyze a large number of posts. This is where AI driven solutions come into play. **Cyberbullying Detection & Behavior Profiling** component is a machine learning based solution that plays a vital role in maintaining a safe environment. It uses NLP techniques, transformer models trained on datasets to analyze comments, posts, messages posted by users and detect negative bullying content and remove them automatically. The system monitors the user behavior overtime based on posting frequency and flags users as harmful users. To ensure transparency, the system utilizes explainable AI tools such as SHAP and LIME. These techniques justify why the posts were flagged. This system protects the online bullied victims by automatically deleting harmful texts and explains why a particular post was marked as harmful. This provides offenders fairness and transparency. Instead of relying on the manual reports it automatically detects such negative content and removes them and it causes to increase the trust of users towards the social media platform.

1.1 Background & Literature Survey

With the increase of technology around the world, cyberbullying may come in various ways. Online harassments, Harmful comments, use of offensive language etc. With new platforms being introduced, they come with certain benefits. However, some users misuse it to bully others. Therefore, the number of cyberbullying victims have increased. As users are exposed to technology at a very young age, they are most likely to face cyberbullying at a very young age and that could even impact users mental well-being in long term. A study shows that "46% of U.S. teens from age 13 to 17 have experienced cyberbullying at least once in a survey conducted on April 14-May 4, 2022." [1]. The studies shows that anxiety, depression, low self-esteem, and, sometimes suicidal tendencies can happen due to a result of cyber bullying [2]. Therefore, detection of such issues have become crucial.

To deal with this issue, during the earlier days, keyword-based detection methods have been used by popular networking sites. The dictionaries were either predefined or user generated. In this approach offensive words are flagged as harmful content. However, these methods have limitations. They fail to recognize indirect bullying that doesn't use harmful words. To detect abusive content in social media, it can be categorized into 4 key approaches: Rule-Based, Machine Learning, Lexicon based Hybrid and Deep Learning [2].

Due to all of these limitations, researches started to inspect on machine learning-based techniques where classification models such as logistic regression, Naïve Bayes were used and trained on labeled datasets where labels categorized as "bullying" or "non-bullying." The studies showed that these approaches capture more better than keyword-based techniques. However, they had difficulties when dealing with unbalanced datasets, and understanding sarcastic context. [3].

Researchers started to explore on feature engineering techniques such as n- grams and part-of-speech tags, with the expansion of natural language processing (NLP). It is said these features works more effectively capturing abusive text. Study shows that to improve detection, applying machine learning by combining textual features with

contextual information to detect harmful comments in You tube [4] .Further researchers have also explored using deep learning approaches for detecting cyberbullying. For example, a study on text classification methods, based on deep learning shows that models such as CNN and LSTM captures semantic and meaning of the context of user generated text [5] . Social media content contains, text, images, videos and emojis. Researchers have explored the integration of multimodal data and some studies have mentioned that on platforms such as Tik Tok and Instagram, use of multi model detection enables to perform better at detection of cyber bullying content [6]. at times cyberbullying is done even without the use of offensive words, in such cases to detect harmful content it is important to use sentiment analysis and emotion recognition. Studies have shown that by integrating deep learning models with sentiment features perform far better in cyber bullying detection tasks [7]. It explains that the understanding of the emotional context and the psychology of the context is important in identifying and detecting harmful content and may result in improving the performance of the system.

Apart from the detection of cyber bullying content, it is important to consider fairness and ethical concerns. For example, by using a bias dataset in training can lead to bias results. Studies shows the importance of considering ethical concerns when detecting cyberbullying, as it build users trust. Techniques such as explainable AI (XAI) are integrated to deliver transparency.

In spite of all the studies and advances, there are many challenges related to that area such as use of imbalanced datasets while training. In the real world as, cyberbullying content are much lesser than normal content, by using such imbalanced data it will cause the models to produce incorrect outputs and it is difficult for the models to learn. And another challenge is identifying indirect bullying in social networks like using different emojis. And it is difficult to adjust and detect content related to various cultures and languages. With the increase of the number of users and posts, there is a high demand for real time detection of cyberbullying.

Overall, studies have shown the various techniques such as integration of multi model approaches, deep learning-based text classification systems, sentiment

analysis and use of explainable AI (XAI) to ensure the fairness and transparency. Both accuracy and resilience have improved due to the developments. However, challenges remain in using bias data, developing real time AI driven solutions, achieving transparency can be explored further in the future.

1.2 Research Gap

With the significant evolvement in technology within the recent past years, Machine learning and Deep learning techniques have been used for detection of cyberbullying. Even though lexicon-based approaches and keyword- based approaches is used for detection, However, when it comes to sarcastic comments, contextual meaning and different languages, they struggled to identify cyberbullying content. Linguistic and sentiment features were incorporated by machine learning classification models such as SVMs and Random Forests and that resulted in an improvement of accuracy and performance. And CNNs, LSTMs, BERT, and Ro-BERTA, transformer-based models have achieved incorporating the newest technology in detecting abusive content [8]. And on platforms such as Instagram and Tik Tok, multi-model analysis has been considered to detect cyberbullying patterns of contents such as metadata, text and images [9] .

In a real world scenario as most content are non-abusive content and abusive content being a lesser, most datasets used are biased towards non abusive content. Hence when training models, they struggle in learning patterns of minority class and the results becomes biased towards the majority class [10]. Additionally, the datasets used for research purposes are limited in size, limited scope mostly coming from one platform. When publicly available datasets are limited in size it is difficult for researches to perform experiments and in such cases the models trained on these datasets will perform well on the same dataset but fail to generalize to new unseen data [11] .

As most of the models are used for English language they provide promising results. However, when it comes to various other languages and cultures, the models struggle to provide effective performance. Present- day systems are not equipped to handle

cyber bullying detection of various cultures and other languages because the way of expressing cyberbullying varies across different cultures [8].

Even though the accuracy of these systems have drastically increased through the years, incorporating fairness and transparency into platforms is still undisclosed. In most models used in systems acts as a black box operation where a reasoning is not provided to why certain predictions were taken [12].

Most studies have been done on static datasets and have not integrated into many social media platforms that works real-time. Even though most studies have done on paper they aren't ready to be incorporated into to real world applications that can handle a massive amount of data [11].

Table 1 Comparison between Research findings and existing systems.

Study	Findings	Research Gap
Research 1: Emmery et al. [11]	Poor generalization of imbalanced datasets.	Data size limitations and data imbalance.
Research 2: Gencoglu. [12]	Improving Fairness with a minimal loss	Limited in use of practical application and intergration of the system.
Research 3: Gitari et al. [3]	Lexicon-based approach was taken for detecting cyberbullying and effectively performed on offensive texts.	Struggled when capturing indirectly offensive, sarcastic context.
Proposed Project	Use of NLP, CNN and Explainable AI. Real time moderation.	Addresses fairness transparency, ethical AI principals, detecting cyber bullying of indirect bullying content (sarcastic comments) and system Integration. Also addresses prevention of Cyberbullying automatically by analyzing harmful behavioral patterns.

1.3 Research Problem

With the evolvement of technology and rice of social media have benefit users globaly by enabling communication facilities. However, with the vast number of users using social media platforms have also increased cyberbullying across social Networks and has caused various negative harmful psychological impact on users. Hence it is essential to detect cyberbullying and prevent it. As it is difficult to perform this operation manually for a massive number of users, this calls for using an automated system that detects harmful content and preventing them from posting by also providing fairness and transparency for such content. This will consider both the victim and the harmful Users aspects.

Despite all the advanced techniques for detecting abusive content, such as the use of deep learning models, sentiment-based analyzing, etc. there are many limitations like dataset imbalances, where models trained on biased datasets results in producing biased outputs towards the majority class as it fails to learn patterns of the minority class [11]. Additionally, lack of transparency and fairness in current systems as most models offer “black box” operations and where no justifications are made on predictions made [12].

In current systems very few systems integrate detection systems into real-time platforms due to various limitations. Many studies have experimented on static datasets and but very few have tested on massive number of data and real time platforms [13], [14].

In summary, The key research problem is that there is no social media platform that automatically detects Cyber bullying platform in real-time and prevent users from uploading such abusive content, while providing fairness and transparency for victims and both harmful users. We can say that accuracy have increased effectively on detection of cyberbullying but still have many shortcomings. When building a trustworthy platform, addressing these issues are important.

2. OBJECTIVES

2.1 Main Objective

The main goal of the component, Cyberbullying Detection & Behavior Profiling of Harmful Users, is to develop a system that can automatically detect Cyberbullying content of user generated content and analyze users' behavior patterns over time and preventing of harmful users from uploading contents such as negative comments, harmful posts etc. The system aims to provide a platform that is emotionally intelligent system. It aims identify abusive texts in posts, comments in posts, also identify users who have been repeatedly bullying users, and uploading such content online. The systems automatically flag such users and limit the actions they can perform. The system will be built and will be incorporating natural language processing (NLP) and Deep learning techniques to identify and detect Cyber bullying content. The system also aims to leverage Explainable AI (XAI) techniques to give reasoning or justify the actions performed by the system. This operation aims to provide user with a transparent and trustworthy social medial platform. This will support both victims and harmful abusers by creating a safer environment, while working align with Ethical AI principals.

2.2. Specific Objectives

- This system aims to leverage NLP and deep learning models to detect abusive contents in posts and texts generated by users.

- The system not only aim to detect but prevent the users from uploading such content by automatically deleting abusive content.

- It aims to do a behavioral analysis on users, track user activities overtime. Identify users who have been abusing and posting harmful content repeatedly overtime.

- By using various techniques such as preprocessing, sampling, augmentation techniques, the system aims to reduce dataset imbalance and improve the performance for harmful content detection.

- Usage of Explainable AI (XAI) to give justifications to why the actions of flagged users are classified as harmful. Aims to incorporate fairness and transparency within the system.

- Integration of both detection of cyberbullying content and behavior analysis into the social media platform that can detect and prevent.

- Evaluate the detection accuracy and improve performance

3. METHODOLOGY

This methodology contains developing of the Cyberbullying detection and harmful user profiling component and integration of the component to the real-time social media platform. It undergoes several stages such as data collection and preprocessing, cyberbullying detection, behavior profiling, system integration, evaluation, and the use of appropriate tools and technologies.

3.1 System Architecture

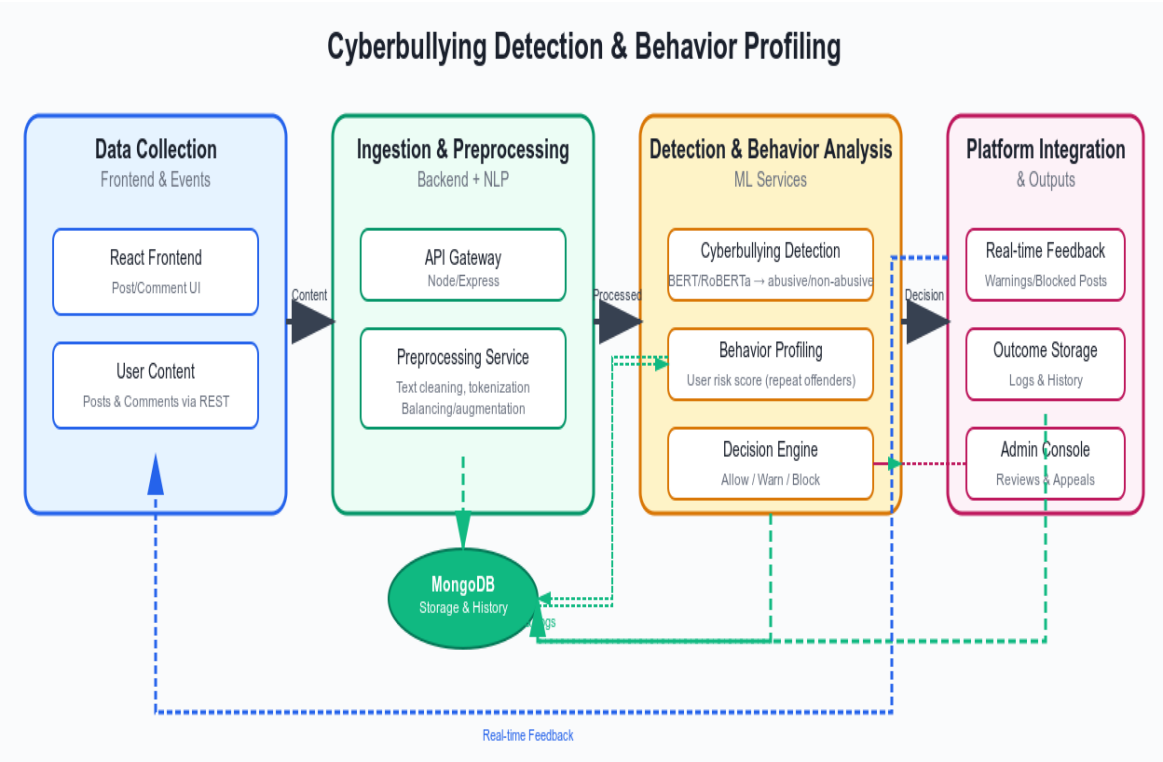

Figure 1 Architecture Diagram of Cyberbullying Detection and Behavior profiling

The above diagram shows the workflow of the Cyberbullying Detection & Behavior Profiling component. User generated content is passed from the frontend for preprocessing, then it is analyzed by the detection module and the behavior analysis module. The system responds by sending warnings, or blocking the content and it is integrated to the platform. MongoDB serves as the main storage unit.

3.2 System Solution

3.2.1 Data Collection and Preprocessing

Dataset Collection:

Datasets will be collected from publicly available repositories and sites such as Kaggle, GitHub and datasets which were used for previous studies. For cyberbullying studies datasets from Twitter, You Tube and Instagram have been generally used for conducting experiments. As the limitation of dataset size may cause data imbalances, multiple datasets will be combined and trained.

Preprocessing:

Text cleaning will be performed under various tasks such as normalization, removing of stop words, lowercasing.

The tokenization or lemmatization will be done by breaking down the text into small tokens converting them to their root forms making easier for the models to adapt and learn.

As Imbalanced datasets are one of the reasons for the models to give incorrect inputs, data imbalances will be handling during this stage. Different techniques will be used to mitigate the biasedness within the dataset. Such as using methods such as SMOTE (Synthetic Minority Oversampling Technique), under sampling of majority classes and oversampling of minority classes.

To overcome the shortcomings of data imbalances, data augmentation techniques such as Contextual Word Embedding where using BERT like models to replace contextually similar words. This supports in overcoming the data imbalance problem.

The dataset will be separated as training dataset where generally a 80% of the dataset is used for training the model and testing dataset where 20% of data will be used for testing purposes

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3.2.2 Cyberbullying Detection Module

Model Selection:

For detection of Cyberbullying content, the Deep Learning models such as Long Short -Term Memory (LSTM), Convolutional Neural Networks (CNN) and transformer-based model like Bert RoBERTA will be used.

Feature Engineering:

Sentiment scores and Word embeddings such as Word2Vec, GloVec will be incorporated to capture the emotional tone and the semantic context of the content.

Training and Optimization:

After splitting the dataset, it will be trained on a balanced dataset with various optimization methods. The Optimization technique Adam will be sued for optimization and regularization techniques to prevent overfitting. To maximize F1 scores and accuracy scores, hyperparameter tuning will also be performed.

Explainability:

Explainable AI (XAI) methods such as SHAP or LIME will be used to highlight key words to provide justification for a detection decision. This ensures transparency of the system. For example: A flagged content may show highlighted terms that triggered it to be flagged.

3.2.3 Behavior Profiling of Harmful Users

Analysis:

This module detects user behavior over time by tracking user actions. By analyzing the user behavior. If the user is classified as a harmful user, users' profile will be flagged and some actions may be limited.

Behavior Metrics: The user profiling will be done based on various user aspects such as posting frequency and severity. If a user is engaging in uploading such content repeatedly over time, they will be flagged as Repeat offenders.

3.2.4 System Integration

Integration into Social Media Platform:

The component of Cyberbullying Detection & Behavior Profiling of Harmful users will be integrated to the systems backend. When a post or a comment is uploaded to the frontend by Users, it will be passed to the backend for analyzing and the detection system will be performing its operations in real-time.

APIs and Data Pipelines:

RESTful APIs will be handling the communication between the detection task and the frontend user Interface.

Transparency and User Feedback:

When a user generated content, if the content is classified as an abusive content it will be flagged and the user will not be allowed to upload the content. But to ensure transparency, the system will justify to why their content was flagged by showing highlighted key words and features.

Scalability Consideration:

The Node.js architecture is efficient in handling high volumes of requests, which enables the system to easily process a massive number of posts.

3.2.5 Evaluation Metrics

- To evaluate the detection performance, F1- score, Accuracy, Precision, and recall scores will be used.
- Confusion matrix will be used for visualization of errors such as false positives.
- Studies and analysis will be conducted to understand and evaluate the explainability of the system. Whether it gives a trustworthy and transparent environment.

3.2.6 Tools and Technologies

Platform Development (MERN Stack)

- **MongoDB:** The User generated content such as posts and comments will be saved using this NoSQL database.
- **Express.js:** This will handle the backend and is responsible in handling the communication between the social media platform, Cyberbullying detection module, frontend and the database.
- **React.js:** This Handles the frontend operations and it can be w used to design and develop a user-friendly UI for users to perform various tasks, such as uploading posts and posting comments on user content.
- **Node.js**: Handles API requests and integrating detection models where it is the Server-side runtime environment

Cyberbullying Detection & NLP

Python: this is the language used for developing the machine learning based module and it is integrated with REST APIs. For the development of this component, various python and machine learning libraries will be used.

- **TensorFlow / PyTorch** for incorporating Deep learning models such as CNNs.
- **Hugging Face Transformers** for NLP models.
- **NLTK and Spacy** libraries for various preprocessing tasks such as tokenization and lemmatization.
- **Pandas, Numpy** for feature Engineering and Data Cleaning.
- **Scikitt-learn** for preprocessing and model evaluation.
- **Matplotlib / Seaborn** for visualization and model analysis.

3.3 Work Break Down structure

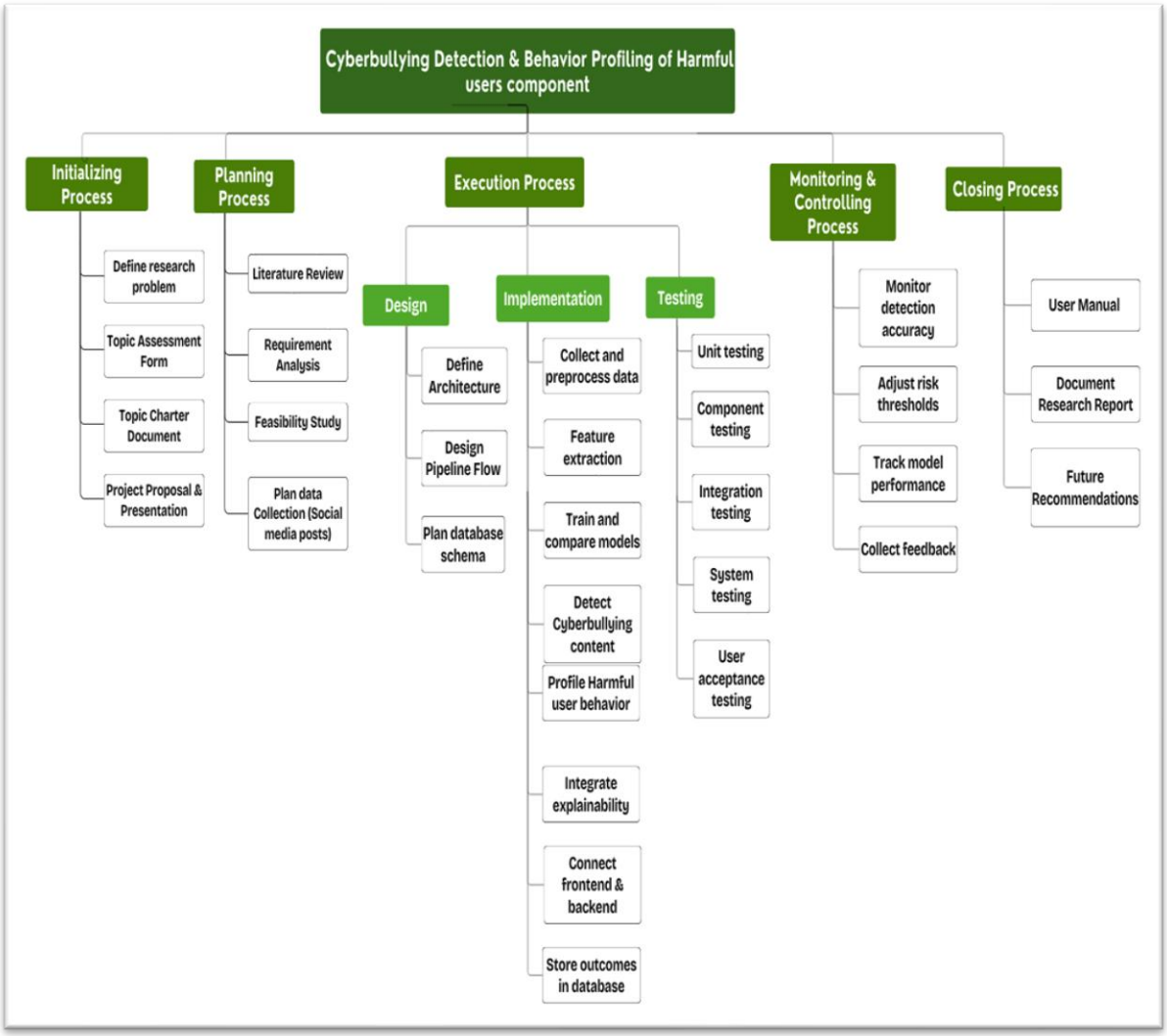

Figure 2 Work Break Down Structure

3.4 Gantt Chart

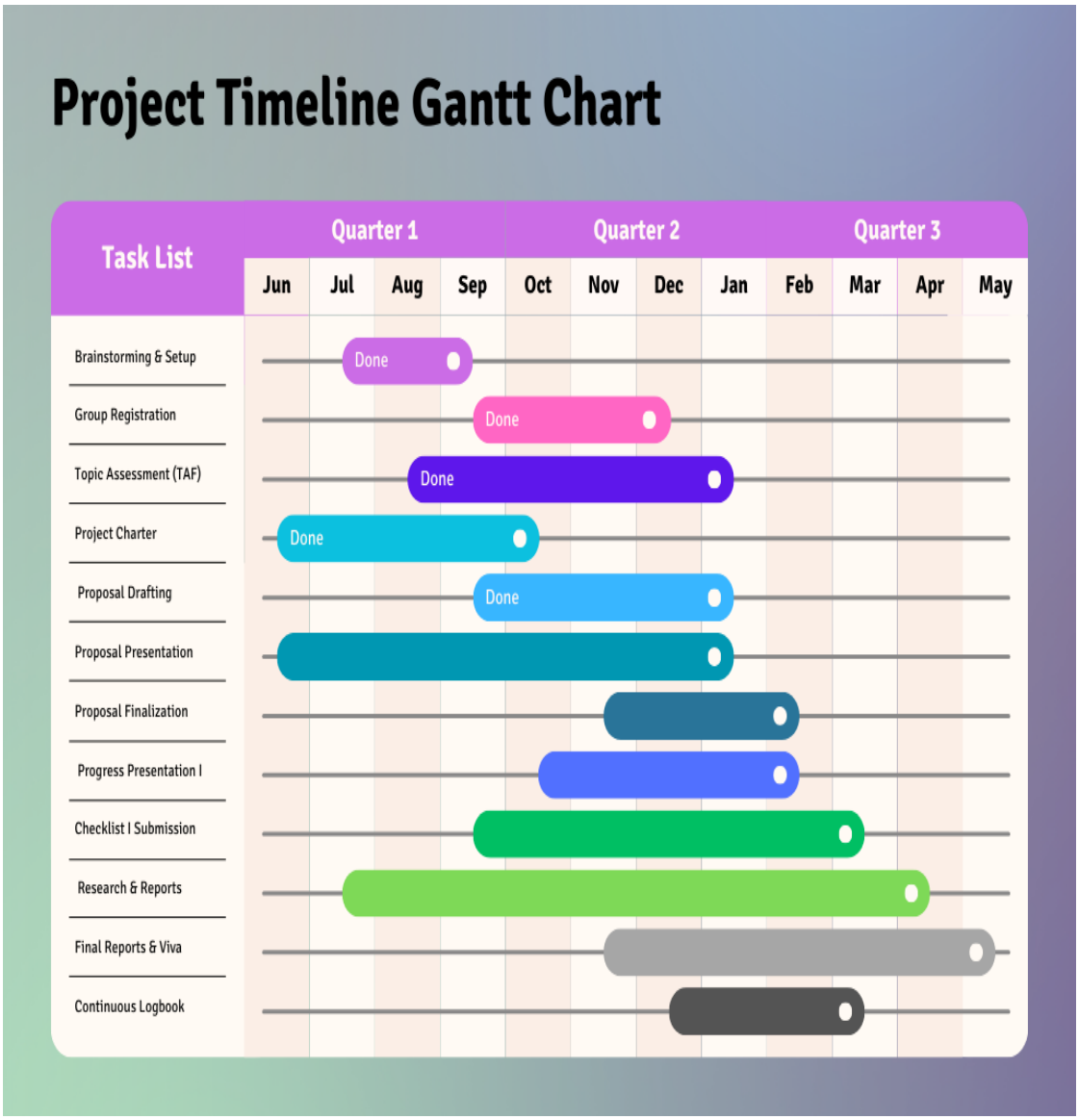

Figure 3 Gantt Chart

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

- The social platform must allow users to upload posts and comments.
- The detection component must automatically detect Cyberbullying content of user generated content.
- The Users must be able to see uploaded posts and contents by other people.
- The prevention of Abusive content being posted. – Whenever a user post a harmful or abusive post, the system must detect and prevent it from being posted.
- The flagging of harmful users or content and the users must be given a reasoning to why their contents were flagged by highlighting abusive terms and phrases.
- There must be a history of flagged users and content.

4.2 Non-Functional Requirements

- **Performance:** The system must process user content to be performed as real-time interactions with less delay.
- The system must support real-time interactions by processing and classifying posts with low latency.
- **Reliability:** The detection component must provide accurate outputs by reducing the false outputs.
- **Fairness:** When detection the system must mitigate bias and have an equal accuracy among various data.
- **Transparency:** Providing correct reasoning to why contents were flagged.
- **Availability:** The system must be accessible for users all the time and must minimize the downtime.
- **Usability:** The system must be user friendly and easy to use.
- **Security:** Users data must be securely stored and transmitted with the use of user authentication.

4.3 Expected Test Cases

The system must be evaluated and tested in multiple levels to ensure that it works functionally correct. Therefore, the testing of accuracy, reliability and fairness etc. plays an important role. Must ensure that this component satisfies all the requirements expected.

Testing Accuracy of the Models

The dataset will be split into, Training Dataset where 70% of the whole dataset will be used for training the model and 15% data will be used for testing of the model and 15% of the dataset used for validation of data. The model performance and reliability will be evaluated by using necessary performing matrices such as precision, recall, F1-score, and confusion matrices.

Unit Testing

Unit testing ensures the correctness of each function in isolation before integration. It will be performed when data is being preprocessed, classification of text and behavior profiling on individual modules.

Integration Testing

This testing method will focus on how the interaction between the detection module and the social media platform communicates. It contains testing the connection and the communication between the frontend React and backend Node.js and the database connection (MongoDB) with the cyberbullying detection module.

System Testing

The entire component is tested to ensure that the system may perform accurately and process large number of posts in real time, analyze user behavior. It also checks the performance, availability and transparency of the deployed system.

User Acceptance Testing (UAT)

Ensuring that the system meets the end users' expected requirements and provide the client with a practical, trustworthy and an easy to access environment.

5. COMMERCIALIZATION AND BUDGETING

Commercialization Strategy: The system can be integrated into existing social media platforms or specialized community networks.

Cost Allocation:

- **Labor:** Development and testing.
- **Cloud Storage:** Google Cloud (100GB × 6 months) ≈ LKR 7,200.
- **Additional Costs:** maintenance cost, user support

Target Market:

- Social media Platforms.
- Educational networking platforms

Value Proposition:
The solution provides:

- Transparent and explainable AI decision-making.
- Ethical and fair recommendations.
- Emotionally intelligent environment that value user well-being.

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